

Profile Summary:

Results-driven Data Scientist with **8.5** years of experience and specializing in developing advanced Machine Learning models. Proficient in Python and R, with a strong foundation in statistics and data analysis. Skilled in extracting meaningful insights from large datasets and translating complex findings into actionable insights for non-technical stakeholders. Expertise in Machine Learning, statistical analysis, and Data Science to drive data-driven decision-making and business impact.

Technical Skills**Programming & Data Handling:**

- Languages: Python, SQL, PySpark
- Data Engineering & Processing: Data Preprocessing, Data Augmentation, Exploratory Data Analysis (EDA), Data Visualization

Machine Learning & Deep Learning:

- Algorithms: Linear, Non-Linear, and Ensemble-based Regression & Classification models
- Statistical Analysis: Statistical Significance Analysis, Anomaly Detection, Feature Engineering, Feature Selection, Feature Extraction, Dimensionality Reduction (PCA)
- Natural Language Processing (NLP): Text Cleaning, Text Classification
- Computer Vision: Image Processing, Image Classification, Object Detection, Image Segmentation, Object Tracking
- Neural Networks: Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), Long Short-Term Memory (LSTM) using TensorFlow-Keras and PyTorch

Cloud Computing & Back-end Development:

- AWS: AWS DynamoDB, S3, Web Sockets, API Gateway, Lambda, EC2, ECR, ECS
- Azure: Azure ML Studio, Web App Service, Container Instances, DataBricks, Blob Storage, Azure SQL, Virtual Machines

DevOps & Deployment:

- CI/CD Pipelines: GitHub Actions, Automated CI/CD Pipelines
- Containerization: Docker-based containerized services and microservices

Image & Video Processing Expertise:

- CNN Architectures: Expertise in architectures such as AlexNet, VGGNet, ResNet for image classification, object detection, and segmentation using TensorFlow and PyTorch
- Transfer Learning: Experience in leveraging pre-trained CNN models (e.g., ImageNet) for domain-specific applications
- Object Detection Techniques: Proficient in R-CNN, Faster R-CNN, and YOLO (You Only Look Once), including anchor-based and anchor-free methods
- GPU Acceleration: Hands-on experience with CUDA for accelerated deep learning computations
- Software Libraries: Proficiency in OpenCV, NumPy, Scikit-Image, PIL, Imutils, and deep learning frameworks like TensorFlow, Keras, and PyTorch

Professional Experience:**JPM Chase, Plano, Tx - Senior Data Scientist****[Aug 2021 - Present] - 3.5 Years**

- Collaborated with stakeholders to gather, analyze, and define project requirements.
- Designed scalable, robust, and efficient architectures tailored to specific business needs.
- Trained and deployed machine learning models using industry-standard platforms and technologies.
- Implemented MLOps best practices, leveraging MLflow and AWS SageMaker for model lifecycle management.
- Developed and maintained RESTful APIs and GraphQL endpoints for seamless data exchange.
- Integrated real-time processing protocols to enhance system responsiveness.
- Managed metadata, insights, and structured/unstructured data using databases such as MySQL, PostgreSQL, and AWS Athena.
- Utilized PySpark for large-scale data processing and AWS Redshift for efficient data warehousing.
- Researched and developed NLP applications, including text classification, named entity recognition (NER), sentiment analysis, question answering, and topic summarization.
- Designed, implemented, and evaluated Proof of Concepts (POCs) and end-to-end NLP solutions using OpenAI language models and open-source LLMs.
- Optimized LLM performance through prompt engineering, Retrieval-Augmented Generation (RAG), and fine-tuning techniques.
- Hands-on expertise with OpenAI, AWS Bedrock, Azure OpenAI, and vector databases for advanced AI solutions.
- Implemented and maintained CI/CD pipelines using Jenkins and Azure DevOps.
- Managed containerized deployments with Docker and Kubernetes for scalable and efficient application hosting.
- Worked closely with cross-functional teams to ensure smooth integration and deployment of solutions.
- Maintained comprehensive documentation of project workflows, processes, and system designs.
- Conducted rigorous testing and validation to ensure model accuracy, reliability, and overall system performance.
- Developed and validated NLP solutions using agile methodologies and best practices.

- Cloud Platforms: AWS (EC2, API Gateway, ECR, SageMaker, Athena, Redshift, Bedrock), Azure (ML Studio, OpenAI)
- Programming & Frameworks: Python, PySpark, SQL, MySQL, PostgreSQL
- DevOps & CI/CD: Docker, Kubernetes, Jenkins, Azure DevOps, Git, GitHub
- AI/ML & NLP Tools: MLflow, TensorFlow, OpenCV, OpenAI, RAG, Vector DB
- Project Management: JIRA, Agile methodologies

Warner Bros, Richardson Texas - Machine Learning Engineer

[October 2019 - July 2021] - 2 years

- Partnered with data engineers, product managers, and developers to scale AI/ML solutions and translate business requirements into actionable data science strategies.
- Developed personalized recommendation systems for Warner Bros streaming platforms using collaborative filtering and deep learning models, increasing user engagement by 25%.
- Built computer vision models for automated scene recognition and metadata tagging, reducing manual video tagging time by 80%.
- Created box office revenue forecasting models, increasing prediction accuracy by 20%, optimizing investment decisions for major theatrical releases.
- Utilized Airflow to extract data from SQL Server and load it into Snowflake, reducing the transactional database load and improving dashboard response time by 60%.
- Implemented AI-driven content moderation using NLP and computer vision techniques to detect policy-violating content, improving compliance automation by 70%.
- Optimized ETL pipelines using Apache Airflow and PySpark, reducing data processing time by 40% and enhancing dashboard performance for executive decision-making.

Cyient, Hyderabad, India - Data Scientist

[Apr 2015 - June 2018] - 3 years

- Collaborated with cross-functional teams to research and build Market Mix Modeling, Multi-Touch Attribution Modeling, and Omni-Channel Effectiveness frameworks, enabling data-driven marketing decisions.
- Designed Built predictive analytics solutions for CPG and retail clients, leveraging regression models and time-series forecasting, leading to a 15% improvement in sales prediction accuracy.
- Conducted hypothesis testing and revenue forecasting, leveraging Causal ML frameworks for CPG and retail clients, resulting in a 15% improvement in uplift measurement accuracy.
- Implemented NLP-based text analytics solutions using TF-IDF, Word2Vec, and LSTMs for customer sentiment analysis and product feedback mining.
- Strengthened brand positioning among industry competitors by leveraging data-driven insights, driving business growth and strategic advantage.
- Applied TensorFlow and PyTorch for machine learning models, focusing on demand forecasting, churn prediction, and recommendation systems.
- Boosted customer experience scores by 20% within 3 months and increased customer retention rates by 5%, demonstrating a significant improvement in customer satisfaction and loyalty.